

RESUME SKILL EXTRACTION USING NATURAL LANGUAGE PROCESSING

Mini Project Report

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1. Abstract

This project focuses on extracting skills from resumes using Natural Language Processing (NLP). Resumes usually contain useful information about a person's technical and professional skills, but finding these skills manually can take time. The aim of this project is to automate this process using a simple NLP-based approach.

For this project, a dataset containing 220 resumes in JSON format was used. The resume text and available skill information were processed and cleaned before performing skill extraction. A rule-based keyword matching method was used to identify skills from the resume text. The extracted skills were then compared with the actual skills available in the dataset.

The performance of the system was evaluated using Precision, Recall, and F1-score. The system achieved a Precision of 52.77%, Recall of 52.58%, and F1-score of 52.68%. The project shows how a simple NLP approach can be used to automatically identify skills from resume content.

2. Problem Statement

Resumes contain important information about a candidate's education, experience, technical knowledge, and skills. However, identifying the relevant skills from a large number of resumes manually can be time-consuming.

The purpose of this project is to develop a simple NLP-based system that can automatically identify skills mentioned in resume text. The extracted skills can then be compared with the skills provided in the dataset to measure how well the system performs.

2.1 Why This Problem Is Relevant

Skill extraction can be useful in applications such as resume screening and recruitment. Automatically identifying skills can help reduce manual effort and make it easier to organize information from resumes. This project demonstrates a basic approach that can be used as a starting point for automated resume analysis.

3. Objectives

- To load and process resume data stored in JSON format.
- To extract useful text and skill information from resumes.
- To preprocess and normalize the skill vocabulary.
- To identify skills from resume text using keyword matching.
- To compare extracted skills with the actual skills in the dataset.
- To evaluate the system using Precision, Recall, and F1-score.
- To present the extracted skills in an understandable format.

4. Dataset Description

The dataset used in this project contains **220 resumes** stored in JSON format. Each JSON file represents one resume and contains information such as the resume ID, resume text, and a list of skills associated with the resume.

The resume text includes information such as work experience, education, technical skills, and other professional details. The skill list provided with each resume was used as the reference or actual skill set for evaluating the extraction system.

The dataset was loaded and processed using Python. The JSON files were read and combined so that the resume text and corresponding skills could be processed together.

```
df.head()
```

	id	text	skills
0	219	Valarmathi Dhandapani\nInvestment Banking Oper...	[Martini (Trading System), IT Support Liaison,...
1	137	Krishna Prasad\nPatna City, Bihar - Email me o...	[Data Entry]
2	71	Sai Patha\nMule ESB Integration Developer - Ci...	[Health sheet generation, Production proactive...
3	99	Vijayalakshmi Govindarajan\nSAP as a Consultan...	[Problem-solving skills, Solution Manager (SAP...
4	117	Divesh Singh\nBengaluru, Karnataka - Email me ...	[Positive Attitude, Hard & Smart Working, Abil...

Sample JSON structure of the resume dataset

5. Dataset Statistics

A total of **220 resumes** were used for the project. Out of these, **219 resumes contained skill information**, while one resume did not contain any skills.

The dataset contained **12,816 total skill entries** and **8,757 unique skills** before normalization. After normalization, the number of unique skills was reduced to **8,258**.

The average number of skills associated with a resume was **58.25**, with the minimum being 0 and the maximum being 284.

```
... Total skill entries: 12816
    Unique skills: 8757
```

Skill statistics before normalization

6. Methodology

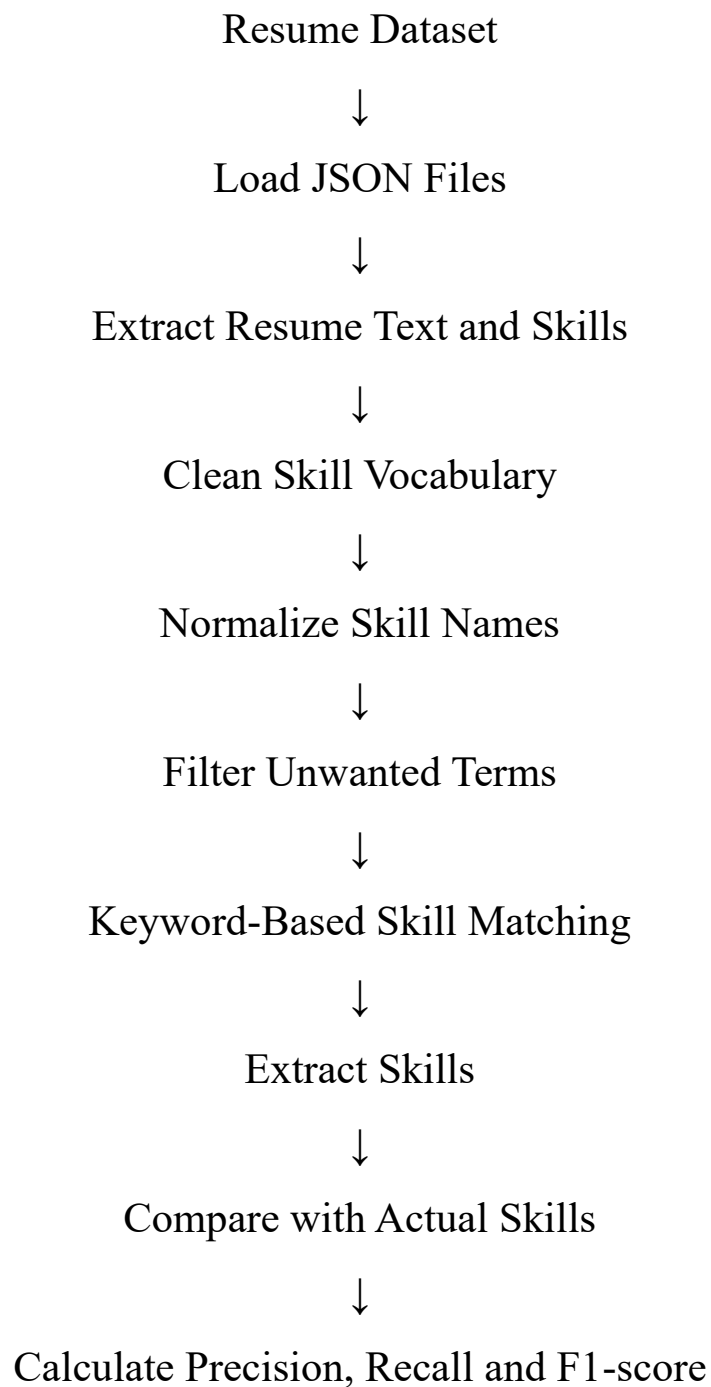
The project uses a **rule-based keyword matching approach** to extract skills from resumes. The main idea is to compare the words and phrases in the resume text with the prepared skill vocabulary.

Before extracting skills, the skill data is cleaned and normalized to reduce differences caused by capitalization and similar variations. Some unsuitable terms are also removed to reduce irrelevant matches.

After preprocessing, the system checks the resume text for matching skills and creates a list of extracted skills for each resume. These

extracted skills are then compared with the actual skills provided in the dataset. The performance of the system is measured using **Precision, Recall, and F1-score**.

6.1 Methodology Workflow



6.2 Skill Preprocessing and Normalization

The skill information in the dataset contains variations in capitalization and formatting. For example, the same skill may appear in different forms such as Excel, EXCEL, and excel. To make the matching process more consistent, the skill vocabulary was normalized before extraction.

After normalization, the number of unique skills decreased from **8,757 to 8,258**. A filtering step was then applied to remove unsuitable or very short skill terms that could lead to irrelevant matches. After filtering, the number of unique skills was **8,236**.

This preprocessing helped create a cleaner skill vocabulary for the keyword-based extraction process.

```
... Original unique skills: 8757  
    Normalized unique skills: 8258
```

Skill vocabulary after normalization

```
Before filtering: 8258  
After filtering: 8236
```

Skill vocabulary after filtering

6.3 Skill Extraction

After preparing the skill vocabulary, the system performs skill extraction from the resume text. Each resume is processed individually, and the text is checked against the prepared list of skills.

When a skill is identified in the resume text, it is added to the extracted skill list for that resume. The extracted skills are stored along with the resume ID so that the results can be analyzed and compared with the actual skills.

For example, when Resume ID **219** was processed, the system identified **80 skills** from the resume text. The extracted skills included

```
... Resume ID: 219

Final Extracted Skills:
['ACCESS', 'AUDITING', 'Accountability', 'Attention to Detail', 'Banking', 'Banking Operations', 'Billing', 'Bio Informatics', 'Bloomberg', 'Budgeting', 'Buffer Utilization', 'CRL Applications',

Number of Skills Extracted: 80
```

The $\mathcal{H}_1$ and $\mathcal{H}_2$ are defined as follows:

The evaluation was performed using three standard measures:

Precision, Recall, and F1-score.

F1 secret:

represents a skill extracted by the system that was not present in the actual skill list. A False Negative represents an actual skill that the system failed to extract.

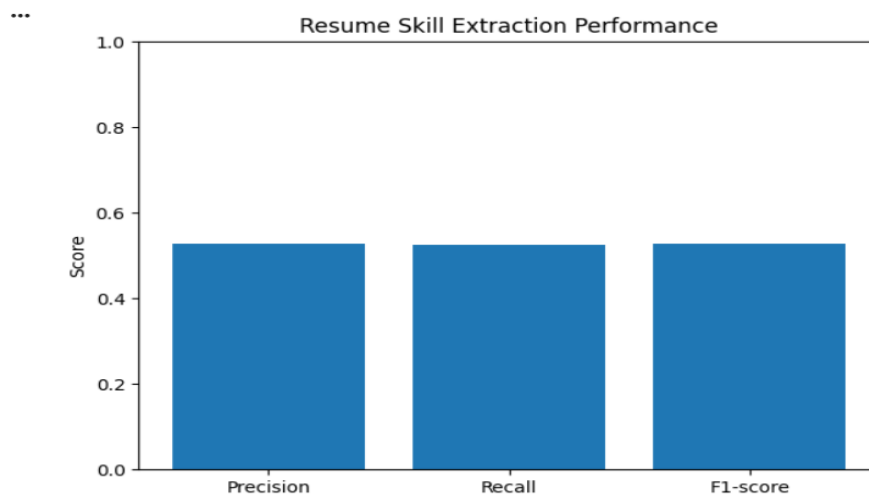
7. Results

The developed system was tested on all **220 resumes** in the dataset. The system successfully generated an extracted skill list for each resume. The extracted skills were then compared with the actual skills available in the dataset to evaluate the performance of the system.

The overall evaluation produced a **Precision of 52.77%**, **Recall of 52.58%**, and an **F1-score of 52.68%**. These results show that the rule-based keyword matching approach was able to identify a reasonable portion of the skills present in the resumes, although some skills were missed and some additional terms were incorrectly identified.

7.1 Overall Performance

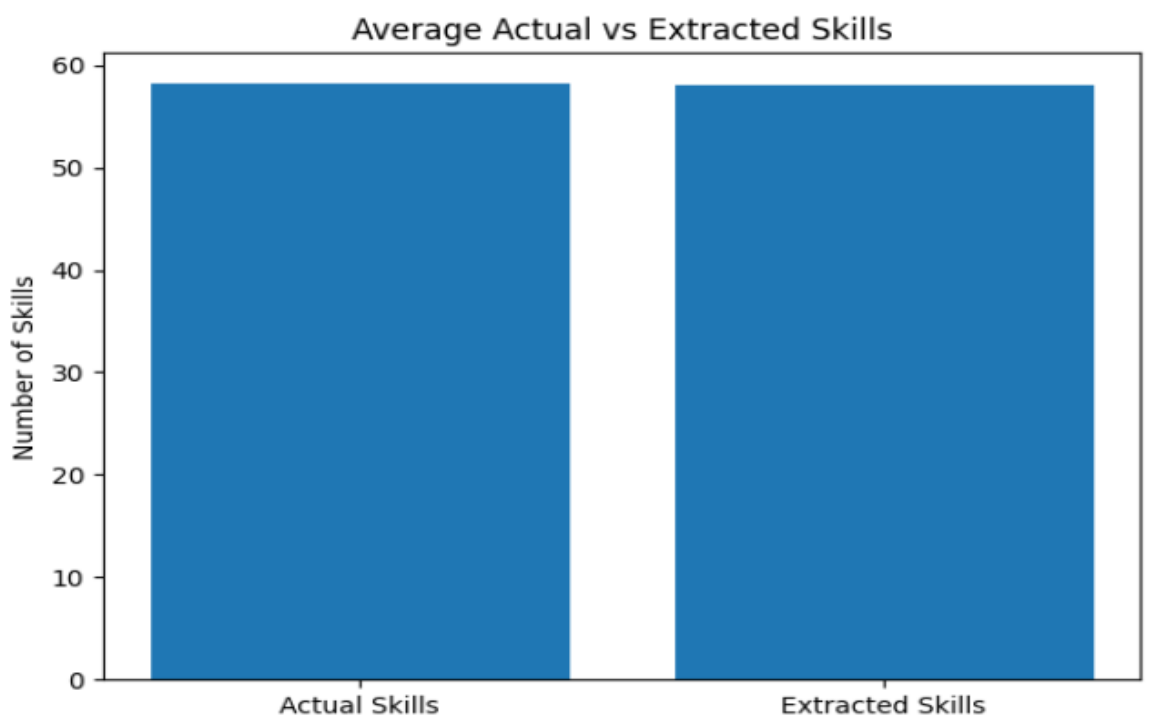
Metric	Score
Precision	52.77%
Recall	52.58%
F1-score	52.68%



Overall performance of the resume skill extraction system

7.2 Skill Extraction Statistics

The system produced an average of **58.03 extracted skills per resume**, while the average number of actual skills was **58.25**. The maximum number of actual skills in a resume was **284**, while the maximum number of extracted skills was **271**.



Comparison of average actual and extracted skills

8. Limitations

Although the system is able to extract skills from resumes, the current approach has some limitations.

First, the extraction depends on the skill vocabulary available in the dataset. If a skill is not present in the prepared vocabulary, the system may not identify it even when it appears in the resume.

Second, different ways of writing the same skill can make matching difficult. Although normalization was applied, some variations in wording may still result in missed skills.

The system can also identify some general words or terms that may not represent actual professional skills. This can result in false positive predictions.

Finally, the current approach is based mainly on keyword matching and does not deeply understand the context in which a word or phrase is used.

9. Future Scope

The current system provides a basic approach for extracting skills from resumes, but it can be improved further with more advanced NLP techniques.

In the future, machine learning and deep learning methods can be explored to understand the context of skills instead of depending mainly on exact keyword matching. Similarity-based techniques can also help identify skills when they are written in different forms.

The system could also be extended to recognize additional resume information such as job roles, education, years of experience, certifications, and projects. This could make the system more useful for automated resume screening and candidate analysis.

Another possible improvement is to develop a user-friendly application where a user can upload a resume and immediately receive the extracted skills and other relevant information.

10. Conclusion

This project demonstrated how Natural Language Processing can be used to extract skills from resume text. A dataset of 220 resumes was processed using a rule-based keyword matching approach.

The project included skill preprocessing, normalization, filtering, extraction, and evaluation. The extracted skills were compared with the actual skills in the dataset to measure the performance of the system.

The system achieved an overall **Precision of 52.77%**, **Recall of 52.58%**, and **F1-score of 52.68%**. The results show that the approach can identify many relevant skills from resume text, while also highlighting the limitations of relying mainly on keyword matching.

Overall, the project provided practical experience in applying NLP techniques to a real-world problem and showed how automated skill extraction can support resume analysis and screening.

11. References

1. Python Documentation — Python programming language and standard libraries.
2. Pandas Documentation — Data loading, processing, and DataFrame operations.
3. Scikit-learn Documentation — Evaluation metrics and machine learning utilities.
4. Natural Language Processing concepts used for text preprocessing and keyword extraction.

- Resume dataset used for the implementation and evaluation of the project.

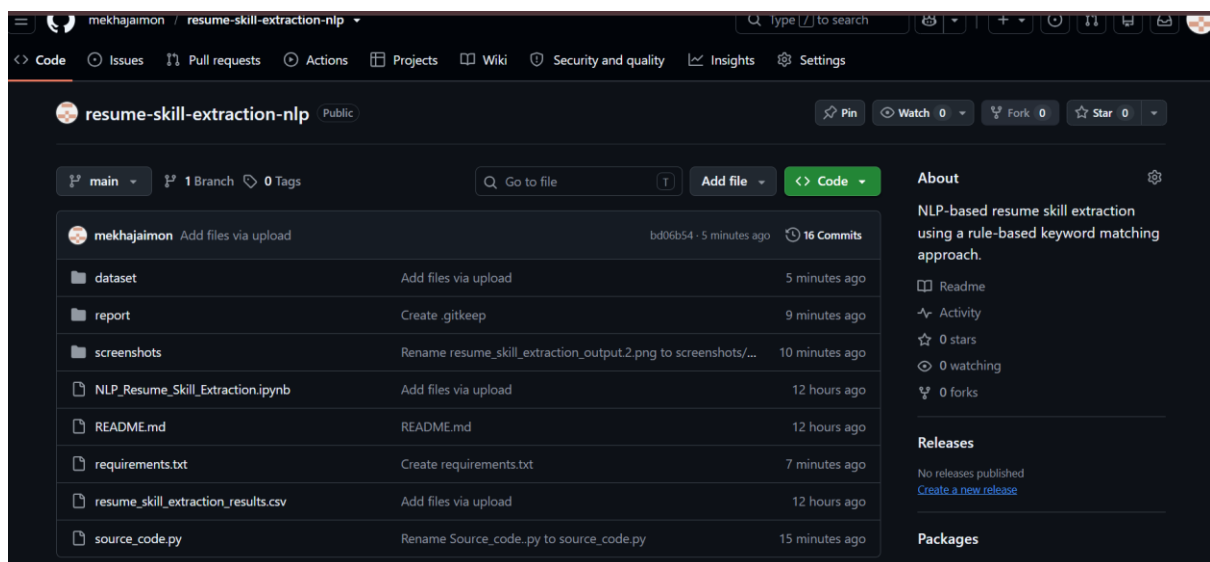
12. GitHub Repository

The complete project files have been organized and uploaded to a GitHub repository. The repository contains the source code, dataset, screenshots, project notebook, README file, extracted results, requirements file, and project documentation.

The repository is organized into separate files and folders to make the project easy to understand and reproduce. The `source_code.py` file contains the main implementation, while the dataset, screenshots, and report folders contain the supporting project materials.

GitHub Repository:

<https://github.com/mekhajaimon/resume-skill-extraction-nlp>



GitHub repository structure of the NLP Resume Skill Extraction project